

# CURRICULUM VITAE

## DO NHU TAI



### PERSONAL INFORMATION

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| Name:     | <b>DO NHU TAI</b>                                                                                                                                                                                     | Birthday:    | January 30th, 1983 |
| Email:    | <a href="mailto:donhutai@gmail.com">donhutai@gmail.com</a>                                                                                                                                            | Phone:       | (82) 010-9128-9618 |
| Homepage: | <a href="http://dntai.vneasy.net">http://dntai.vneasy.net</a><br><a href="http://github.com/dntai">http://github.com/dntai</a><br><a href="https://leetcode.com/dntai">https://leetcode.com/dntai</a> | Nationality: | Vietnamese         |
| Contacts: | Skype: nhutai                                                                                                                                                                                         |              |                    |

### WORKPLACES

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| Current work:      | Postdoc Researcher                                                                                                                                 |
| Time Period:       | From 08/2017 – current                                                                                                                             |
| Workplace:         | Department of Artificial Intelligent Convergence, Chonnam National University, 77 Yongbong-ro, Yongbong-dong, Buk-gu (61186), Gwangju, South Korea |
| Advisor Professor: | Prof. Kim Soo-Hyung                                                                                                                                |
| Laboratory:        | Pattern Recognition Lab - Room 331, Building 7                                                                                                     |
| Website:           | <a href="https://prlabjnu.github.io/">https://prlabjnu.github.io/</a>                                                                              |

### RESEARCH INTERESTS

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- Pattern Recognition, Computer Vision, Medical Image Analysis, Deep Learning,
- Emotion Recognition, Hand Gesture Recognition, Knee Bone Tumor Detection, Face Tracking, Video Understanding

### EDUCATION

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2017 – 08/2021: Department of Artificial Intelligent Convergence, Chonnam National University, South Korea (<http://jnu.ac.kr>)

- Major: Artificial Intelligent Convergence in Doctor of Philosophy in Engineering with 36/36 credits and GPA 4.5/4.5.
- Graduation thesis: “*Facial Emotion Recognition Using Context-Aware Spatio-Temporal and Temporal-Pyramid Models*”, advised by Prof. Kim Soo-Hyung, defended in June 2021.

2013 – 2016: Faculty of Computer Science and Engineering, International University, Vietnam National University, Ho Chi Minh City (<http://www.hcmiu.edu.vn>)

- Certificate: Master of Engineering in Information Technology Management (GPA in scale 100: 84.8 – Very Good)
- Graduation thesis: “*A Speeded-up temporal median filter and its application for background modelling in video surveillance*”, advised by Dr. Le Thanh Sach, defended on March 2016 at 3B Ly Tu Trong, Ben Nghe Ward, District 1, HCMC

2001 – 2005: Faculty of Information Technology, Ho Chi Minh City University of Foreign Languages - Information Technology (<http://www.huflit.edu.vn>)

- Certificate: Bachelor of Information Technology (GPA in scale 10: 8.71 – Very Good)
- Graduation thesis: *Applying wavelet in image and video compression problems*, advised by Dr. Vu Thanh Hien, defended on August 2005 at 155B, Su Van Hanh Street, Ward 13, District 10, HCMC.

## WORKING EXPERIENCE

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08/2021 – Current: Postdoc Researcher at Pattern Recognition Lab, Department of Artificial Intelligent Convergence, Chonnam National University, South Korea.

2017 – 08/2021: Ph.D. Candidate and Research Assistant, Department of Artificial Intelligent Convergence, Chonnam National University, South Korea.

2005 – 08/2017: Lecturer, Faculty of Information Technology, Ho Chi Minh City University of Foreign Languages - Information Technology (HUFLIT) (<http://www.huflit.edu.vn>).

## AWARDS

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### Awards in Postdoc Fellowship at Chonnam National University

- Winner of 2021 Korean Emotion Recognition Challenge, <https://www.kaggle.com/c/kercc2021/leaderboard>, Team ADLER
- Winner of 2021 Aquaculture Artificial Intelligence Idea Contest, <http://sarc.jnu.ac.kr/contest/20211105/>, Team ADLER

### Awards in Ph.D studying at Chonnam National University

- Best Presentation Award:
  - Tram-Tran Nguyen-Quynh, **Nhu-Tai Do**, Soo-Hyung Kim, "**MLEU: Multi-Level Embedding U-Net for Fully Automatic Image Colorization**," Proc. Int. Conf. Machine Learning and Smart Computing(ICMLSC 2020), pp. 119-123, Haiphong City, Vietnam, Jan. 2020. (BRL) (Best Presentation Award)
- Best Paper Award:
  - **Nhu-Tai Do**, Tram-Tran Nguyen-Quynh, Thanh-Hung Vo, 정성택, 김수형, "**Seg-Unet 및 Dice Loss 를 이용한 방사선 사진에서의 End-to end 뼈 종양 분할**," 2019년도 한국컴퓨터종합학술대회 논문집(KCC 2019), pp. 799-801, 제주 ICC, 2019년 6월. (BRL/LINC+) (★우수논문) (End-to-End Knee Bone Tumor Segmentation and Classification from Radiographs using Seg-Unet and Dice Loss)
- Challenges:
  - 2019 Korean Emotion Recognition Challenge, Position 3<sup>rd</sup> <https://www.kaggle.com/c/kercc>
  - Affwild2 Challenge 2019, Position 4<sup>th</sup> <https://ibug.doc.ic.ac.uk/resources/fg-2020-competition-affective-behavior-analysis/> (Team: CNU\_ADL)
  - EmotiW Challenges 2019, Position 6<sup>th</sup> <https://sites.google.com/view/emotiw2019>

## Awards in Vietnam

- Certificate of Merits of Vietnam Union of Science and Technology Associations for supervising students in Vietnam Olympiad Information Technology Student Contest and ACM/ICPC Contest (2015)
- Consolation prize in Student Scientific Research of Ministry of Education and Training (2005) with research “***Applying Wavelet in Image and Video Compression Problems***”.
- 3<sup>rd</sup> prize of super cup block in Vietnam Olympiad Information Technology Student Contest (2005)

## PROJECTS

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01/2020 – current: BRL Project (Korea Basic Research Funding) – Period 2:

- Topic: Multi-Modal Context Emotion Recognition in Video Understanding

12/2019 – current: Medical Prognosis:

- Topic: Knee Bone Tumor on MRI sequences

03/2018 – 01/2019: Medical Project:

- Topic: Knee Bone Tumor Detection from X-ray images

04/2018 - 05/2018: AI Challenge: Fake Face Detection from GAN

- Topic: Fake Face Detection from GAN

10/2017 – 12/2019: BRL Project (Korea Basic Research Funding) - Period 1:

- Topic: Emotion Recognition and 3D Face Emotion
- Topic: Face Tracking
- Topic: AffWild2 Challenge (02-03/2020)
- Topic: Korean Emotion Challenge (KERC) (09-11/2019)
- Topic: EmotiW Challenge (05-07/2019)

09/2017 - 12/2018: Hand Gesture Recognition

- Topic: Hand Gesture Recognition

06/2015 – 12/2016: Graduation thesis “*Speed Up Temporal Median Filter and Its Application in Background Estimation*”

- It is a part of project “Vision-based Traffic Parameter Estimation and Visualization” sponsored by Dept. of Science and Technology in Ho Chi Minh City
- Mainly focus on:
  - The proposed thesis’s objectives is to speed-up the filter using GPU (Graphics Processing Unit).
  - The speeded-up temporal median filter will be also analyzed for background modelling problem in video surveillance.
  - Expected results of thesis are a new technique for speeding-up the temporal median filter using GPU is proposed and it is faster than recently developed

algorithms, the effectiveness of the proposed temporal median filter is analyzed for the background modelling problem to show features on computation time and on the accuracy.

- Project was implemented in C/C++, OpenCV, MFC and Cuda.

10/2012 – 05/2015: Joining Computer Vision Group of Faculty of Computer Science and Engineering, International University, Vietnam National University, Ho Chi Minh City

- Research GPU algorithms for Image/Video Processing using CUDA.
- Research Numerical methods, Image/Video Processing algorithms
- Support Cuda algorithm for paper “*Subsequent Processing of Background Modeling for Traffic Surveillance System*”.

10/2004 – 08/2005: Bachelor thesis “*Applying Wavelet in Image and Video Compression Problems*”

- The objective of the thesis focused on the rate, quality compression capabilities along with scalability in image and video compression compared to JPEG and MPEG compression standard. The approach of the project is based on wavelet theory, wavelet transform applied to the problem of image compression, video and several algorithms used to quantize the wavelet coefficients as SPIHT, CSPIHT.
- Project was implemented in C/C++, MFC.

## PATENT

| International Patent |                                                                                                                                                                                                                                                                           |
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| Korean Patent        |                                                                                                                                                                                                                                                                           |
| 2021<br>(Review)     | Kim Soo-hyung, Do Nhu Tai, Yang Hyung-Jeong, Oh A-ran, <b><i>Knee bone tumor detection system and method using X-ray image</i></b> , Application No.: 10-2021-0083706 (2021.06.28) (Physicologist) <a href="#">Link</a>                                                   |
| 2021<br>(2022)       | Kim Soo-hyung, Tran Nguyen Quynh Tram, <b><i>Do Nhu Tai</i></b> , O Aran, <b><i>Black and white image automatic coloring system and method</i></b> , Application number: 10-2021-0059216 (2021.05.07). Registration number: 10-2467062 (2022.11.09.) <a href="#">Link</a> |

## PUBLICATIONS

| Current working |                                                                                                                                                                                                                                                                          |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |                                                                                                                                                                                                                                                                          |
| Review          | Trong-Nghia Nguyen; Bo-Gun Kho; <b><i>Nhu-Tai Do</i></b> ; Ngumimi-Karen Iyortsuun; Guee-Sang Lee; Hyung-Jeong Yang, <b><i>Temporal Variational Autoencoder Model for In-hospital Clinical Emergency Prediction</i></b> , Biomedical Signal Processing and Control, 2022 |
| Review          | <b><i>Nhu-Tai Do</i></b> , Minh-Tran Hai, Tram-Tran Nguyen-Quynh, Soo-Hyung Kim, <b><i>DDA: Dual-Domain Attention for Facial Expression Recognition</i></b> , ICIP, 2023                                                                                                 |
| Review          | <b><i>Nhu-Tai Do</i></b> , Hoang-Son Vo-Thanh, Tram-Tran Nguyen-Quynh, Soo-Hyung Kim, <b><i>3D-DDA: 3D Dual-Domain Attention for Brain Tumor Segmentation</i></b> , ICIP, 2023                                                                                           |
|                 |                                                                                                                                                                                                                                                                          |

| International (SCI) Journals |                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021                         | <b>Nhu-Tai Do</b> , Sung-Taek Jung, Hyung-Jeong Yang, Soo-Hyung Kim, “ <b>Multi-Level Seg-Unet Model with Global and Patch-Based X-Ray Images</b> ,” <i>Diagnostics</i> , 2020                                                                                                                                                                               |
| 2021                         | <b>Nhu-Tai Do</b> , Soo-Hyung Kim, Hyung-Jeong Yang, Guee-Sang Lee, Soonja Yeom, "Context-aware Emotion Recognition in the wild using Spatio-temporal and Temporal-pyramid Models," <i>Sensors</i> , 21, 2344, pp. 1-29, March 2021. (IF: 3.275) (BRL2/지역우수)                                                                                                 |
| 2020                         | Tram-Tran Nguyen-Quynh, Soo-Hyung Kim, <b>Nhu-Tai Do</b> , "Image Colorization Using the Global Scene-Context Style and Pixel-Wise Semantic Segmentation," <i>IEEE Access</i> , Vol. 8, pp. 214098-214114, Nov. 2020. (IF: 3.745, NRF2/BRL2)                                                                                                                 |
| 2020                         | <b>Nhu-Tai Do</b> , Soo-Hyung Kim, Hyung-Jeong Yang, and Guee-Sang Lee, "Robust Hand Shape Features for Dynamic Hand Gesture Recognition using Multi-Level Feature LSTM," <i>Applied Science</i> , 10(18), 6293, pp. 1-28, doi:10.3390/app10186293, Sep. 2020. (IF: 2.474) (NRF2/BRL2)                                                                       |
| 2019                         | <b>Do Nhu Tai</b> , In Seop Na, Soo Hyung Kim, "HSFE Network and Fusion Model based Dynamic Hand Gesture Recognition," <i>KSII Trans. Internet and Information Systems</i> , Vol. 14, No. 9, pp. 3924-3940, Sep. 2019. (IF 0.648, BRL)                                                                                                                       |
| 2019                         | Son Thai Ly, <b>Nhu-Tai Do</b> , Soo-Hyung Kim, Hyung-Jeong Yang, Guee-Sang Lee, "A novel 2D and 3D multimodal approach for in-the-wild facial expression recognition," <i>Image and Vision Computing</i> , 92, 103817, pp. 1-12, DOI: 10.1016/j.imavis.2019.10.003, Dec. 2019. (IF: 3.103, BRL)                                                             |
| International Conferences    |                                                                                                                                                                                                                                                                                                                                                              |
| 2021                         | <b>Nhu-Tai Do</b> , Hyung-Jeong Yang, Guee-Sang Lee and Soo-Hyung Kim, "Facial Emotion Recognition in video clips using 3D Face Reconstruction," Proc. 2021 Int. Conf. Smart Media and Applications, Gunsan, Korea, pp. 156-160, Sep. 2021.                                                                                                                  |
| 2021                         | <b>Nhu-Tai Do</b> , Sung-Taek Jung, Soo-Hyung Kim, "Boosting up knee bone tumor detection from radiology and magnetic resonance imaging by using deep learning techniques," Proc. AACR Virtual Special Conf. Artificial Intelligence, Diagnosis, and Imaging, Jan. 2021.                                                                                     |
| 2020                         | <b>Nhu-Tai Do</b> , Tram-Tran Nguyen-Quynh, Soo-Hyung Kim, "Affective Expression Analysis in-the-Wild using Multi-Task Temporal Statistical Deep Learning Model," Proc. 15th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2020), doi: 10.1109/FG47880.2020.00093, pp. 624-628, Buenos Aires, Argentina, Nov. 2020. (BRL/지역우수) |
| 2020                         | Tram-Tran Nguyen-Quynh, <b>Nhu-Tai Do</b> , Soo-Hyung Kim, "MLEU: Multi-Level Embedding U-Net for Fully Automatic Image Colorization," Proc. Int. Conf. Machine Learning and Smart Computing (ICMLSC 2020), pp. 119-123, Haiphong City, Vietnam, Jan. 2020. (BRL) ( <b>Best Presentation Award</b> )                                                         |
| 2019                         | Son Thai Ly, <b>Nhu-Tai Do</b> , Guee-Sang Lee, Soo-Hyung Kim, Hyung-Jeong Yang, "A 3D Face Modeling Approach for In-the-wild Facial Expression Recognition on Image Datasets," Proc. 2019 IEEE Int. Conf. Image Processing (ICIP 2019), pp. 3492-3496, Taipei, Taiwan, Sep. 2019. (BRL)                                                                     |
| 2019                         | Son Thai Ly <b>Nhu-Tai Do</b> , Guee-Sang Lee, Soo-Hyung Kim, Hyung-Jeong Yang, "Multimodal 2D and 3D for In-the-wild Facial Expression Recognition," Proc. 2019 IEEE CVPR                                                                                                                                                                                   |

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|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | Precognition Workshop, Long Beach, USA, June 2019. (BRL)                                                                                                                                                                                                                                                                                                                          |
| 2019                                    | <b>Nhu-Tai Do</b> , Sang-Don Joo, Hyung-Jeong Yang, Sung Taek Jung and Soo-Hyung Kim, " <b>Knee Bone Tumor Segmentation from radiographs using Seg-Unet with Dice Loss</b> ," Proc. Int. Workshop on Frontiers in Computer Vision (IWFCV 2019), Gangnung, Korea, Feb. 2019. (LINC+)                                                                                               |
| 2018                                    | <b>Nhu-Tai Do</b> , In-Seop Na, Hyung-Jeong Yang, Guee-Sang Lee, Soo-Hyung Kim, " <b>Forensics Face Detection from GANs Using Convolutional Neural Network</b> ," Proc. 2018 Int. Symp. Information Technology Convergence, pp. 376-379, Jeonju, Korea, Oct. 2018. (BRL/GCS)                                                                                                      |
| 2018                                    | <b>Nhu-Tai Do</b> , In-Seop Na and Soo-Hyung Kim, " <b>Face Tracking with Convolutional Neural Network Heat-Map</b> ," Proc. 2nd Int. Conf. Machine Learning and Soft Computing, pp. 29-33, Phu Quoc, Vietnam, Feb. 2018. (BRL/GCS)                                                                                                                                               |
| <b>국내논문지 (Domestic Journals)</b>        |                                                                                                                                                                                                                                                                                                                                                                                   |
| 2022                                    | Trong-Nghia Nguyen, Soo-Hyung Kim, <b>Nhu-Tai Do</b> , Thai Thi Ngoc Hong, Hyung Jeong Yang, Guee Sang Lee, " <b>A TabNet-Based System for Water Quality Prediction in Aquaculture</b> ," Smart Media Journal, Vol. 11, Vol. 2 Ho, pp. 39-52, Mar. 2022 ( <b>Winner of 2021 Aquaculture Artificial Intelligence Idea Contest</b> )                                                |
| 2020                                    | <b>도누타이</b> , 정성택, 양형정, 김수형, "다단계 Seg-Unet 모델을 이용한 방사선 사진에서의 End-to-end 골 종양 분할 및 분류," 정보과학회논문지, 제 47 권 제 2 호, pp. 170-179, 2020 년 2 월. (의사과학자)<br><b>Nhu-Tai Do</b> , Sung-Taek Jung, Hyung-Jeong Yang, Soo-Hyung Kim, " <b>End-to-end Bone Tumor Segmentation and Classification from X-ray Images by Using Multi-level Seg-Unet Model</b> ", 정보과학회논문지, pp. 170-179, 2020 년 2 월 |
| 2018                                    | <b>Do Nhu Tai</b> , Soo-Hyung Kim, Guee-Sang Lee, Hyung-Jeong Yang, In-Seop Na, A-Ran Oh, " <b>Tracking by Detection of Multiple Faces using SSD and CNN Features</b> ," 스마트미디어저널, 7 권 4 호, pp. 61-69, 2018 년 12 월. (BRL/지역우수)                                                                                                                                                    |
| <b>학술발표논문집 (Conference Proceedings)</b> |                                                                                                                                                                                                                                                                                                                                                                                   |
| 2020                                    | Tram-Tran Nguyen Quynh, <b>Nhu-Tai Do</b> , 오아란, 김수형, "Generative Adversarial Model 을 이용한 자연경관 이미지의 컨텍스트 색채화", 2020 한국스마트미디어학회 추계학술대회 발표논문집, Vol. 9, No. 2, pp. 86-89, 광주시청자미디어센터, 2020 년 11 월. (지역우수/BRL2)                                                                                                                                                                       |
| 2020                                    | Nguyen-Quynh Tram-Tran, 김수형, <b>Do Nhu-Tai</b> , " <b>U-net 을 사용한 자연 경관 흑백 이미지의 색채화</b> ", 2020 한국스마트미디어학회 춘계학술대회 발표논문집, Vol. 9, No. 1, pp. 84-88, 조선대학교, 2020 년 5 월. (지역우수)<br>(Colorization of Natural Scene Image using U-net)                                                                                                                                                 |
| 2019                                    | <b>Nhu-Tai Do</b> , Tram-Tran Nguyen-Quynh, Thanh-Hung Vo, 정성택, 김수형, " <b>Seg-Unet 및 Dice Loss 를 이용한 방사선 사진에서의 End-to end 뼈 종양 분할</b> ", 2019 년도 한국컴퓨터종합학술대회 논문집(KCC 2019), pp. 799-801, 제주 ICC, 2019 년 6 월. (BRL/LINC+) (★우수논문)<br>(End-to-End Knee Bone Tumor Segmentation and Classification from Radiographs using Seg-Unet and Dice Loss)                                    |
| 2019                                    | <b>Nhu-Tai Do</b> , Son Thai Ly, Guee-Sang Lee, Jung Hee Kim, Soo-Hyung Kim, " <b>A New Approach to Facial Emotion Recognition in-the-wild using 3DFace Reconstruction</b> ,"                                                                                                                                                                                                     |



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|      | 한국스마트미디어학회 2019 년도 춘계학술대회 학술발표논문집, 제 8 권, 제 1 호, pp. 441-442, 한국교통대학교, 2019 년 4 월. (BRL)                                                                                                                                                                                 |
| 2018 | Nhu-Tai Do, Hai-Duong Nguyen, Young Chul Kim, Guee-Sang Lee, Hyung Jeong Yang, Soo-Hyung Kim, " <i>Depth-based Hand Interaction System using Convolution Neural Network</i> ," 한국스마트미디어학회 2018 년도 춘계학술대회 학술발표논문집, 제 7 권, 제 1 호, pp. 98-100, 중앙대학교, 2018 년 4 월. (GCS/BRL) |

## REFERENCES

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### Dr. Tran Thuong Khanh

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